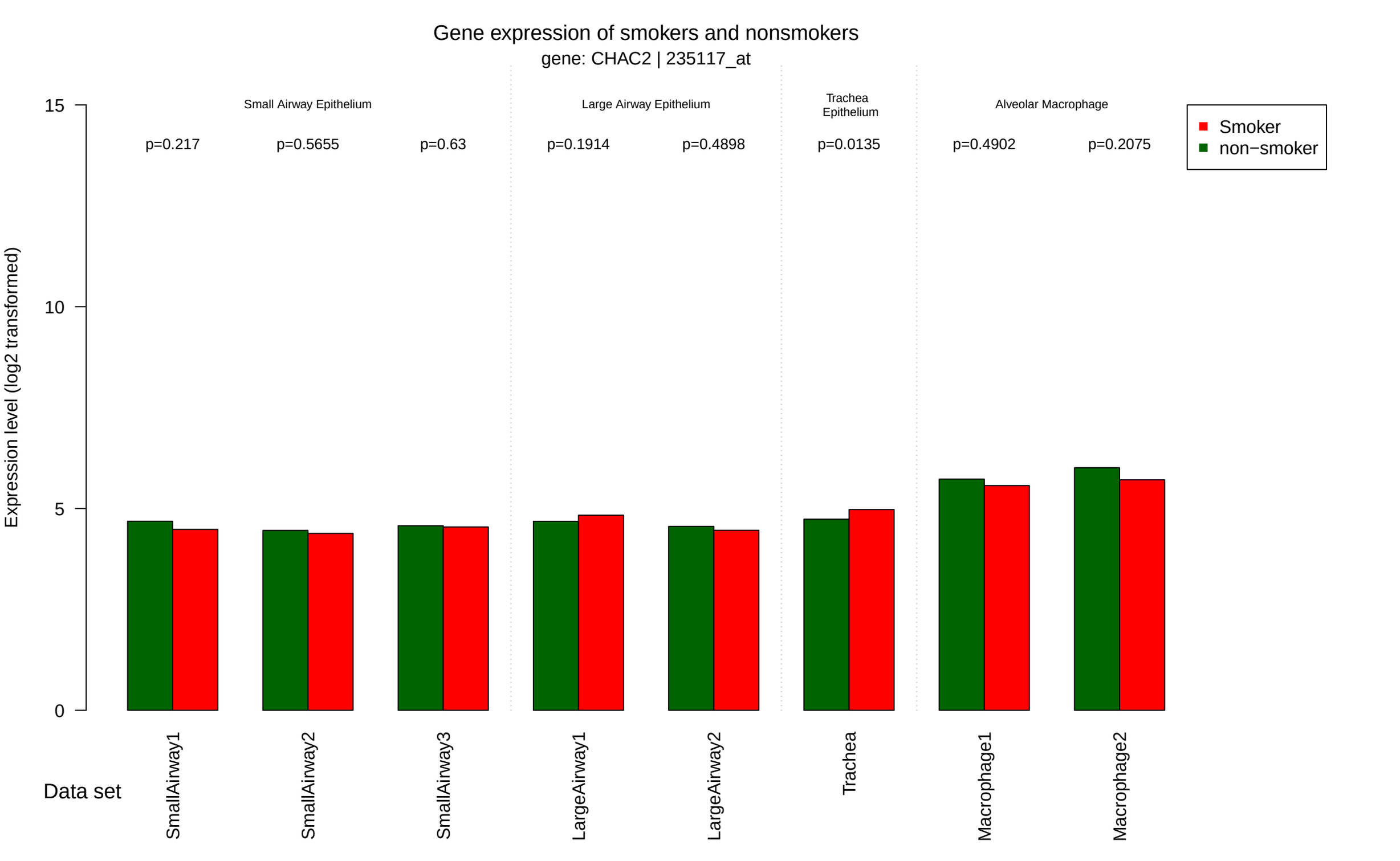
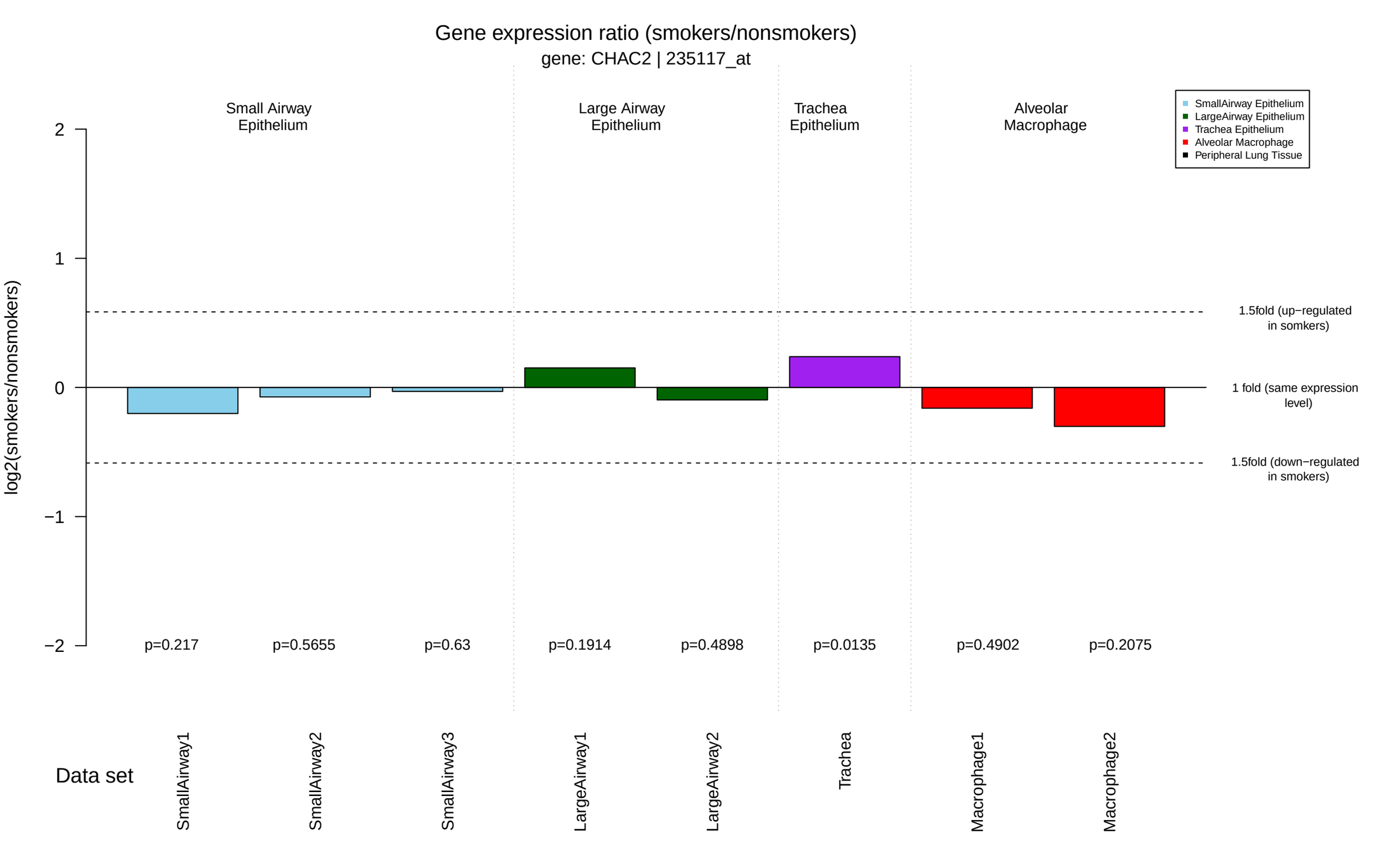
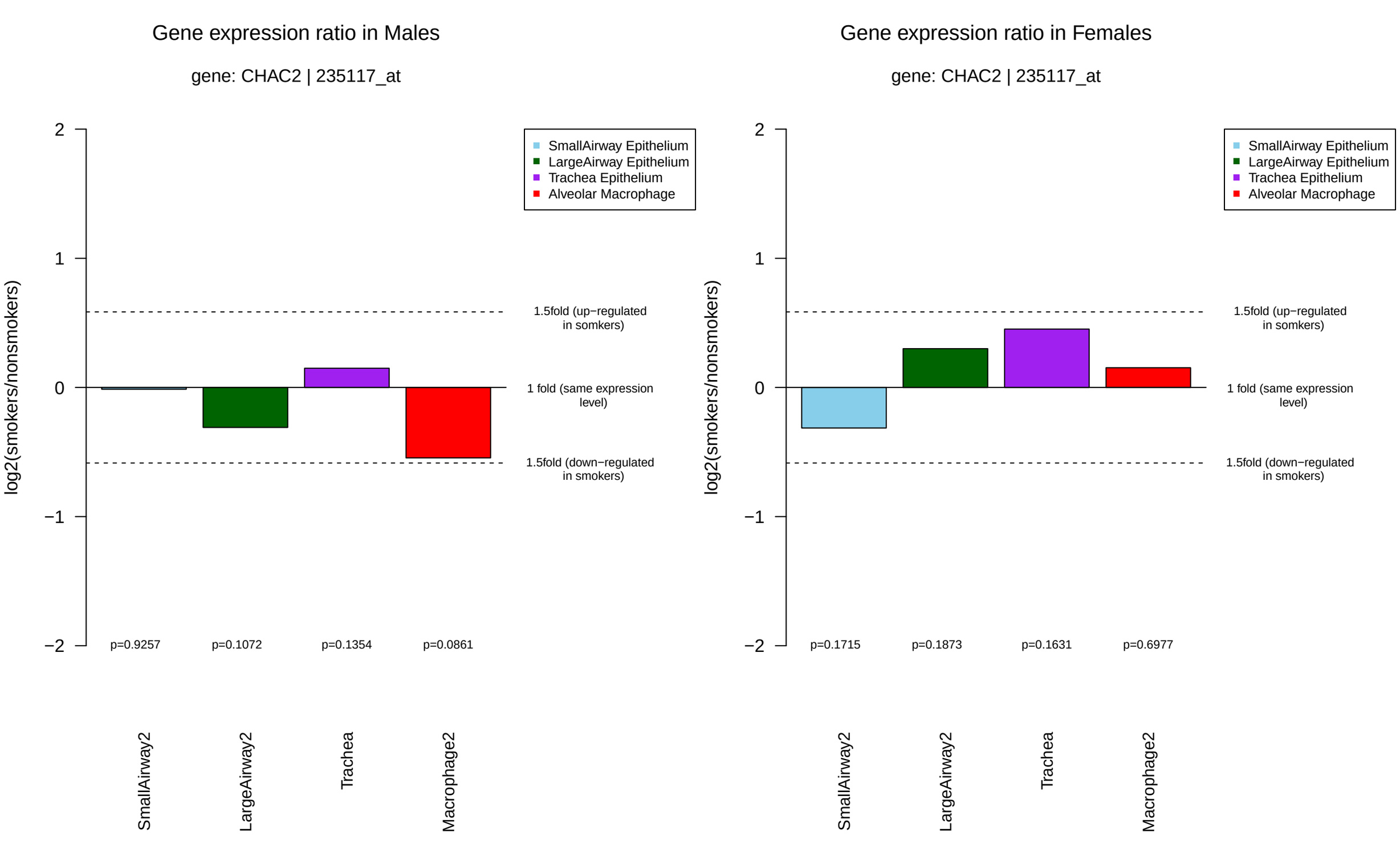
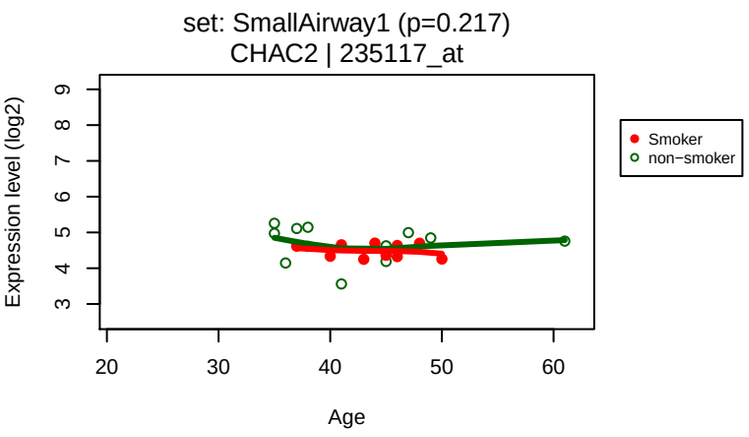
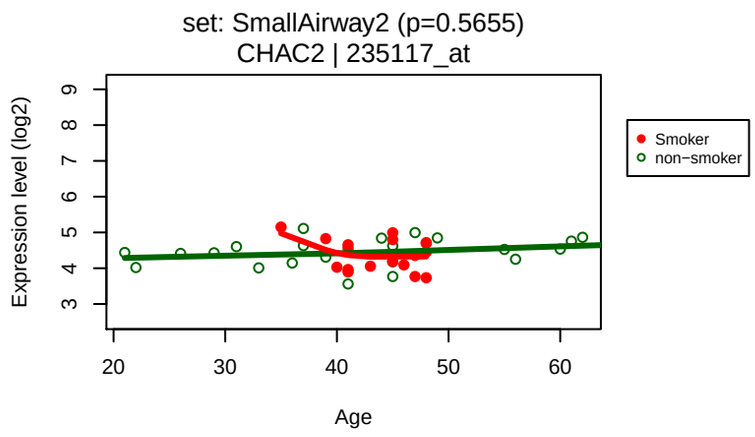
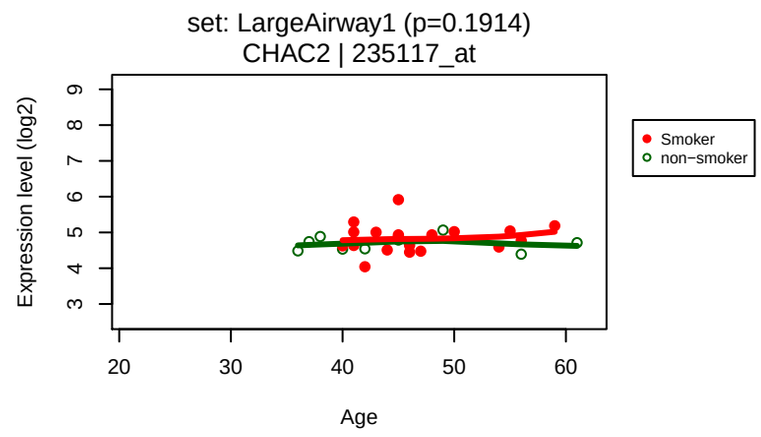

Gene expression of smokers and nonsmokers
gene: CHAC2 | 235117_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.217
p=0.5655
p=0.63
p=0.1914
p=0.4898
p=0.0135
p=0.4902
p=0.2075
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

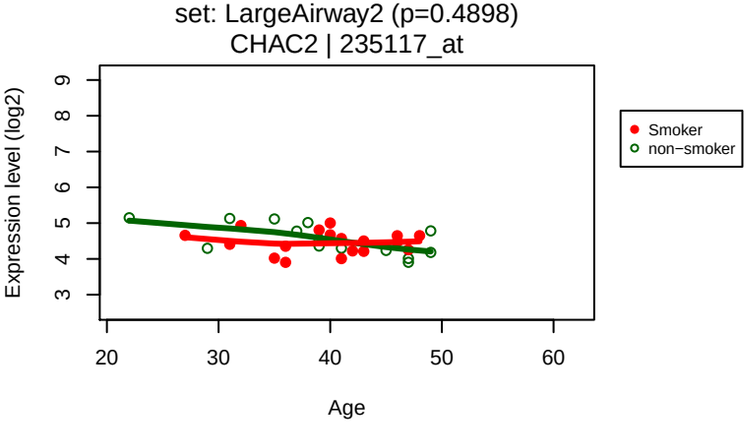

Gene expression ratio (smokers/nonsmokers)
gene: CHAC2 | 235117_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.217
p=0.5655
p=0.63
p=0.1914
p=0.4898
p=0.0135
p=0.4902
p=0.2075
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

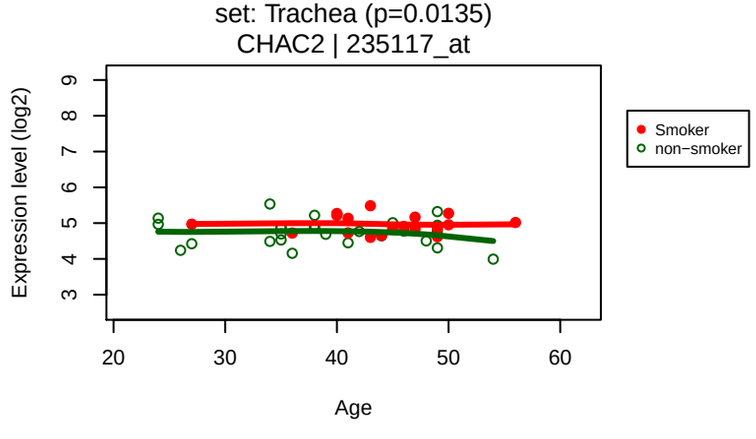

Gene expression ratio in Males
gene: CHAC2 | 235117_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9257
p=0.1072
p=0.1354
p=0.0861
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CHAC2 | 235117_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1715
p=0.1873
p=0.1631
p=0.6977
SmallAirway2
LargeAirway2
Trachea
Macrophage2

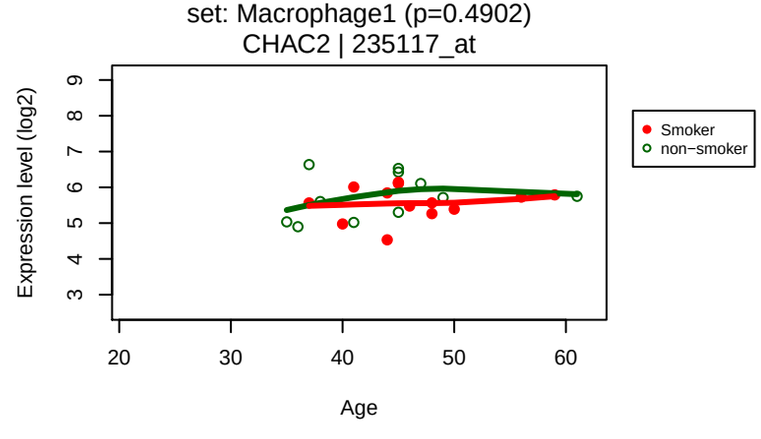

set: SmallAirway1 (p=0.217)
CHAC2 | 235117_at
Expression level (log2)
Age
Smoker
non−smoker

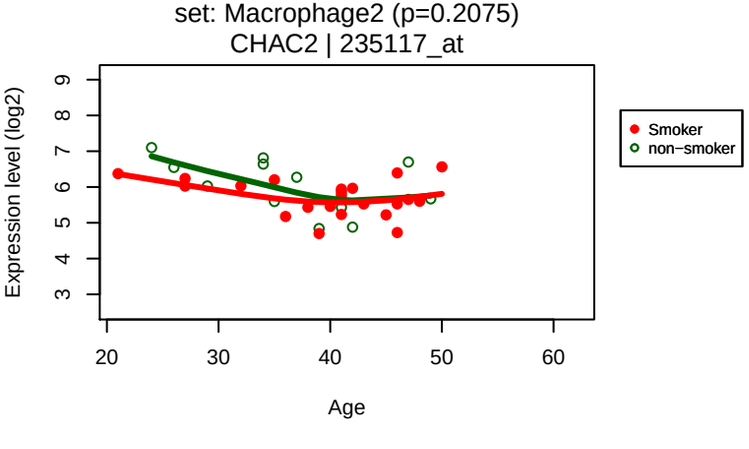

set: SmallAirway2 (p=0.5655)
CHAC2 | 235117_at
Expression level (log2)
Age
Smoker
non−smoker

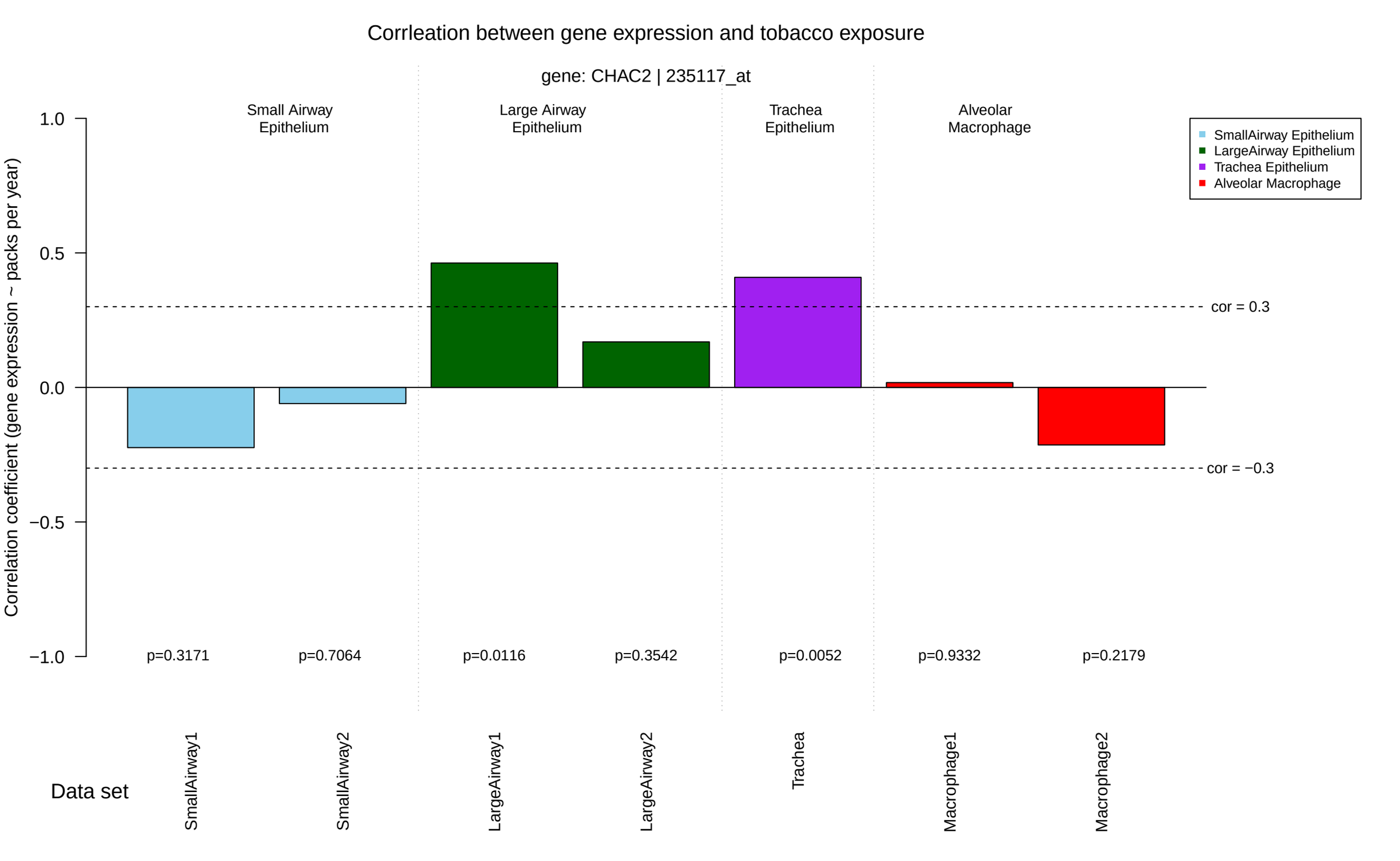

set: LargeAirway1 (p=0.1914)
CHAC2 | 235117_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4898)
CHAC2 | 235117_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0135)
CHAC2 | 235117_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4902)
CHAC2 | 235117_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2075)
CHAC2 | 235117_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CHAC2 | 235117_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3171
p=0.7064
p=0.0116
p=0.3542
p=0.0052
p=0.9332
p=0.2179
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2